Tunnelling Through the Fog for Long Distance Visibility

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1 Introduction

Fog presents a significant challenge to long-range visibility, impacting transportation, surveillance, and communication systems. This project, titled "Tunnelling Through the Fog for Long-Range Visibility," aims to develop a method to locally clear a path through fog by heating and evaporating suspended water droplets and ice crystals through a collimated beam with an appropriately selected radiation spectrum.

To achieve this, the project first investigates the optical properties of fog particles—both liquid droplets and ice crystals—by analysing their complex refractive indices. Using this information, the absorption and scattering coefficients are calculated, along with the scattering phase functions that describe redirection of light due to interactions with these particles. These calculations are essential to quantify the processes of in-scattering and out-scattering, which govern the interaction of radiation with the fog medium.

Building on this foundation, a computational model is developed to simulate the coupled interaction between the fluid flow and the fog particles with collimated beam radiation. The Volume of Fluid (VOF) method is employed to capture the multiphase dynamics, allowing for the detailed modelling of the influence of heated zones on local evaporation or sublimation of fog particles. This integrated approach will help identify the required energy input and delivery strategy to effectively create a clear tunnel through the fog, improving visibility over long distances.

Now, upon heating water particles due to collimated beam radiation, they will vaporise, and ice crystals will melt; hence, it will create a buoyancy effect, and the heated vapour will rise. Subsequent fluid will replace it, which will again be heated, and the process will continue till the fog clears out through the path of the collimated beam. This fluid flow behaviour will be a complex phenomenon that will govern the clearing of the fog for long-range visibility, hence it needs to be studied in depth.

2 Computational Modelling

The Volume of Fluid (VOF) method coupled with collimated beam radiation is employed for the computational modelling of the fog-clearing process. The VOF method is an Euler-Euler approach, which is well-suited for capturing the interface dynamics between two continuous phases — in this case, water droplets (fog) and humid air. This method allows for accurately simulating the multiphase flow behaviour and the interaction between the heated air and the water droplets.

2.1 Governing Equations:

2.1.1 Continuity Equation:

$$\frac{\partial U_i}{\partial x_i} = -\frac{1}{\rho}\frac{d\rho}{dt} \tag{1}$$

2.1.2 Volume Fraction Equation:

$$\frac{\partial \alpha_q}{\partial t} + \frac{\partial}{\partial x_j}(\alpha_q u_j) = \frac{\dot{m}_q''}{\rho_q} \tag{2}$$

where volume fraction is given as

$$\alpha_q = \frac{V_q}{V} \tag{3}$$

The sum of volume fractions for all the phases, q, is given as

$$\sum_q \alpha_q = 1 \tag{4}$$

Also, the properties of the mixture can be calculated as

$$\rho = \sum_q \alpha_q \rho_q, \quad \mu = \sum_q \alpha_q \mu_q, \quad \lambda = \sum_q \lambda_q \mu_q \tag{5}$$

2.1.3 Momentum Equation:

$$\frac{\partial(\rho u_i)}{\partial t} + \frac{\partial}{\partial x_j}(\rho u_i u_j) = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j}\left[\mu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)\right] + \rho g_i + (F_s)_i \tag{6}$$

2.1.4 Energy Equation:

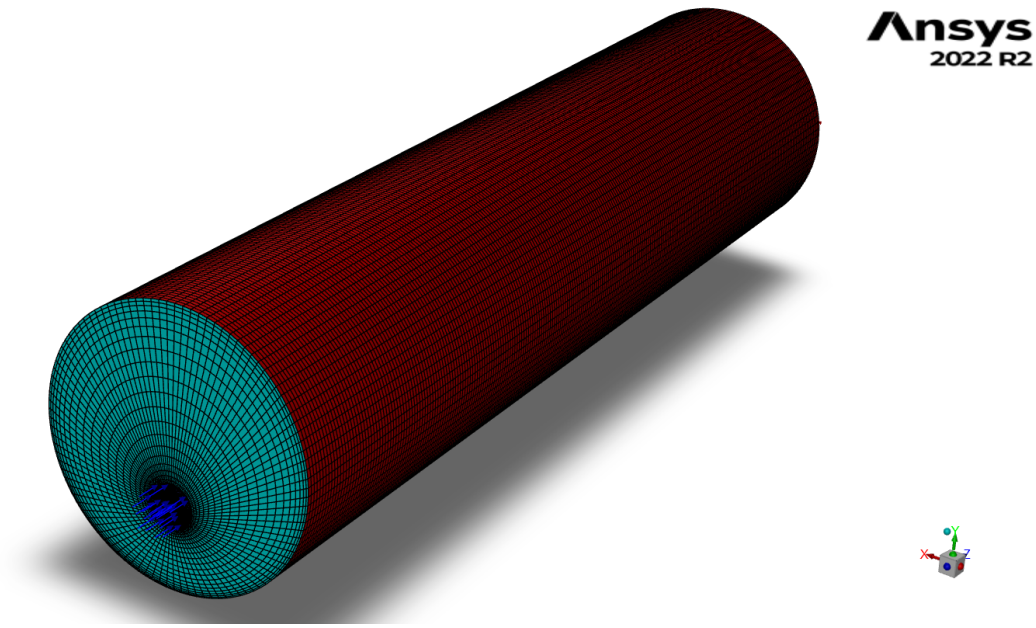
$$\frac{\partial(\rho h)}{\partial t} + \frac{\partial(\rho h u_j)}{\partial x_j} = \lambda\frac{\partial^2 T}{\partial x_j \partial x_j} + q'' + S_R \tag{7}$$

2.1.5 Radiation Modelling:

Monte Carlo method (Statistical Method)

2.2 Geometry for simulation:

For the initial simulations, the computational domain is designed as a cylindrical geometry representing the region where the collimated beam of the selected radiation spectrum interacts with the fog. The geometry has the following specifications:



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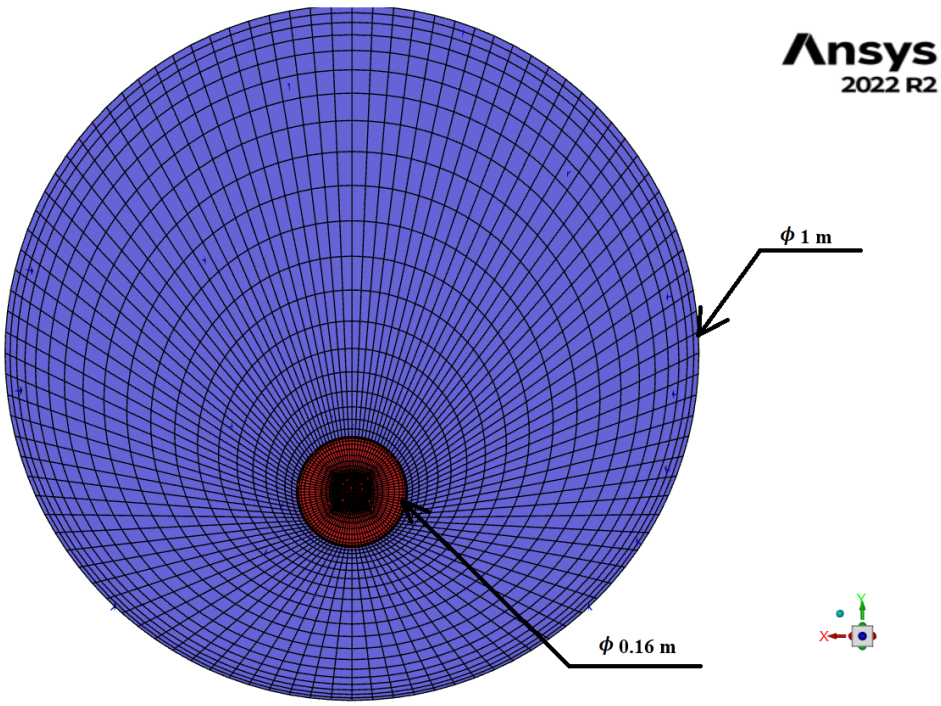

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Figure 1: Computational Domain: (a) isometric view (b) Front Face

The overall environment of the computational domain is filled with the fog (humid air, water droplets & ice crystals mixture), and a collimated beam with selected radiation spectrum is passed through the computational domain through the inner core to heat and evaporate or sublimate the fog particles. This geometry provides a simplified but representative setup to study the interaction between the radiation beam, fluid flow, and fog particles before scaling to more complex or real-world geometries.

3 Optical Properties of Water/Ice:

Water droplets in fog typically range in size from 1 μ to 20 μ, while ice crystals can have diameters from 20 μ to 50 μ. To transfer energy through radiation to such particles, infrared (IR) and microwave radiation are often considered. Infrared radiation spans wavelengths from approximately 0.78 μ to 1000 μ, while microwave radiation covers a broader range, from about 1000 μ to 1,000,000 μ (1 m). For Mie theory to be applicable, the size parameter x = $\frac{2\pi r}{\lambda}$ must be greater than 1. This condition is generally satisfied in the infrared region, where the wavelength is comparable to or smaller than the particle size, making Mie theory valid. However, in the microwave region, the wavelengths are much larger than the particles, leading to a very small size parameter $x \ll 1$. In this case, Mie theory is no longer applicable, and the Rayleigh approximation becomes valid. In the Rayleigh regime, scattering is weak and nearly isotropic.

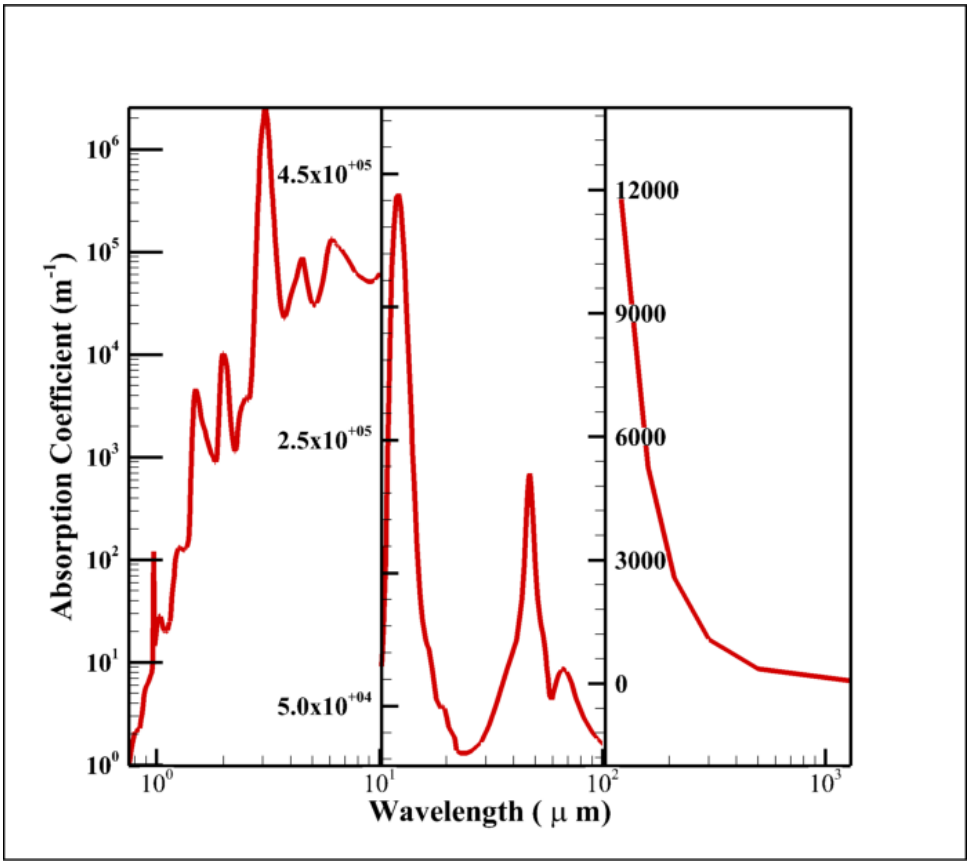
However, it is important to note that absorption can still be significant in this regime. For example, water strongly absorbs microwave radiation, as used in microwave ovens, where rotational modes of water molecules resonate with microwave frequencies, leading to efficient dielectric heating. Thus, even when scattering is negligible, strong absorption can lead to effective radiative heating in the Rayleigh regime, especially for materials like water.

3.1 absorption Coefficient:

The infrared range exhibits the highest absorption coefficients for both water and ice; however, a closer look at the spectrum reveals that the near-infrared range (approximately 1 μ to 10 μ shows particularly strong absorption, especially for ice. Therefore, instead of considering the entire infrared spectrum, this narrower band may be more effective for radiative energy transfer to ice particles.

In contrast, water shows its peak absorption around 1000 μ (far-infrared/microwave). However, the variation in absorption coefficient across the spectrum for water is relatively moderate compared to that of ice. Hence, when both ice crystals and water droplets are present in the medium, it may be more practical to focus on wavelengths that are effective for both.

Additionally, using ice properties as the primary design criteria could be more critical, since ice particles are typically larger and require more energy to melt. Targeting the lower end of the infrared spectrum (e.g., 1–10 μ, where ice absorption is strong) may thus be a more effective strategy for radiative heating.



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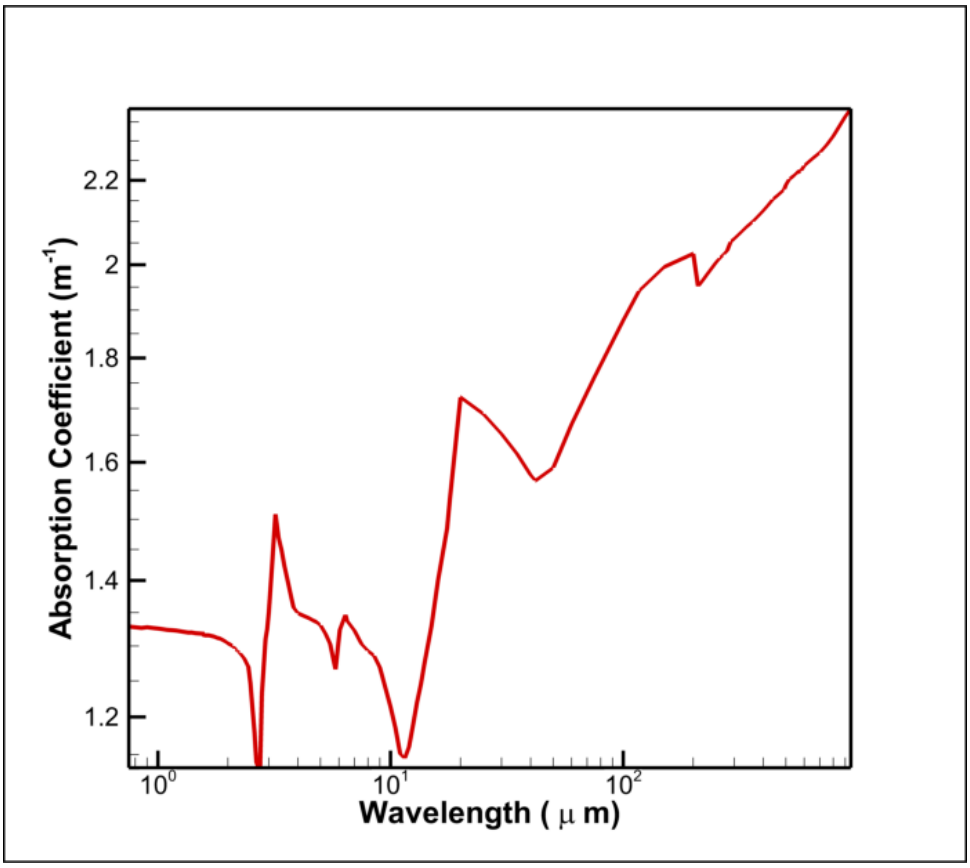

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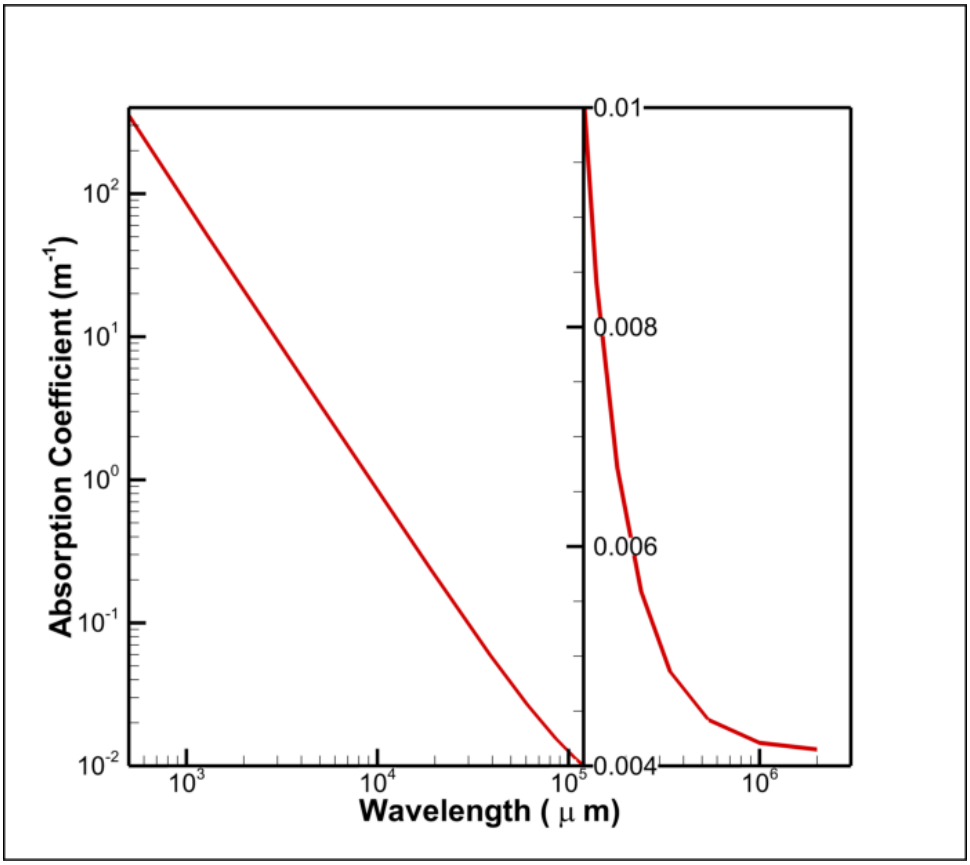
Figure 2: Absorption Coefficient of Infrared range: (a) Ice crystals (b) Water droplets

In the microwave range, water exhibits a peak absorption toward the higher end of the spectrum, specifically between 1000 μ and 10,000 μ. Similarly, ice also shows a peak near 1000 μ, followed by a gradual linear decay in absorption coefficient with increasing wavelength. Based on this behavior, the 1000–10,000 μ range appears to be a reasonable candidate for focused study in radiative energy transfer involving both water and ice particles.

However, it is important to note that this range falls within the Rayleigh regime, where the particle size is much smaller than the wavelength ($x \ll 1$). This brings unique challenges and requires careful theoretical and computational treatment. In the Rayleigh regime, scattering is weak and nearly isotropic, but absorption mechanisms dominate, especially for polar materials like water. Therefore, further investigation is essential to accurately model and understand the energy transfer processes in this spectral range.



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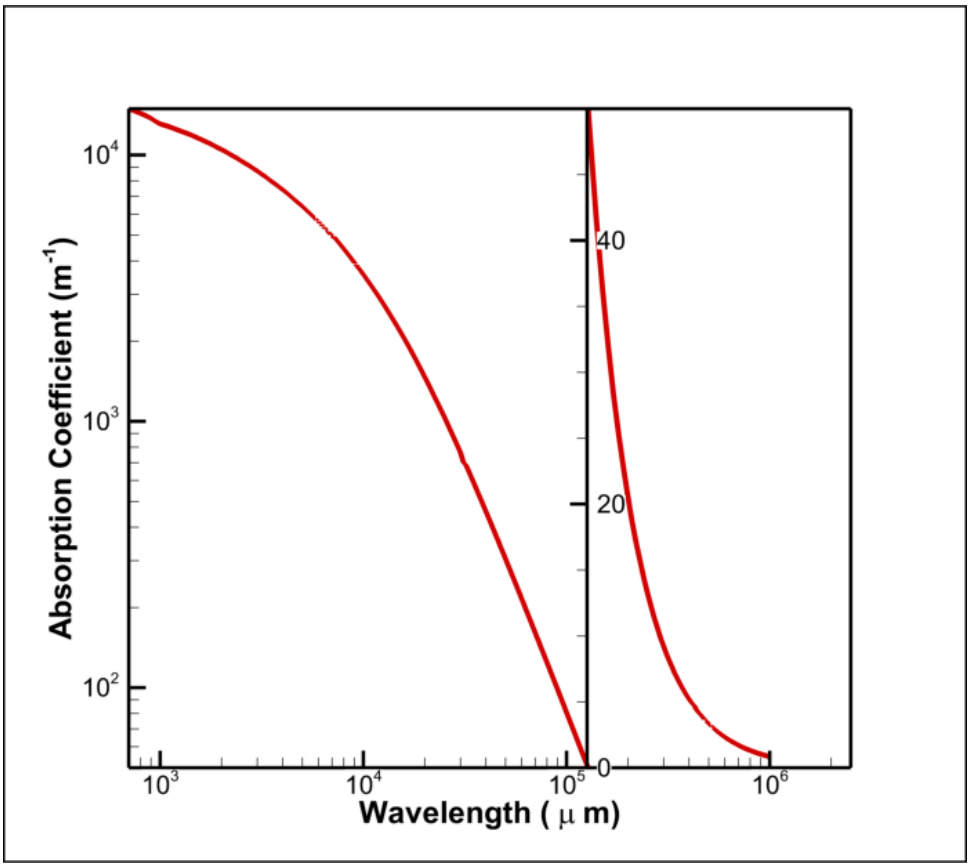

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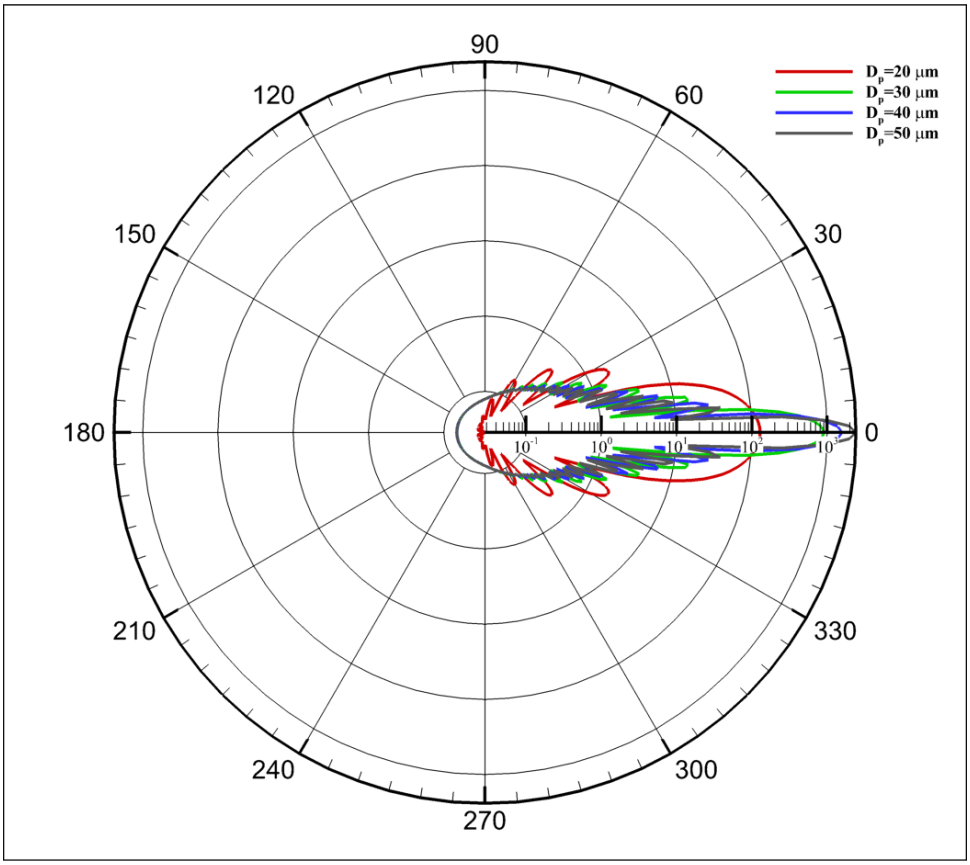
Figure 3: Absorption Coefficient of microwave range: (a) Ice crystals (b) Water droplets

3.2 Scattering Phase Function of Water/Ice:

The scattering phase function is a critical parameter that describes how radiation is redirected after interacting with a particle. Specifically, it indicates the angular distribution of scattered light, helping us understand whether the scattering is predominantly forward, backward, or isotropic. Forward scattering means that after the interaction, the radiation continues in roughly the same direction as the incident beam, whereas in backward scattering, it is redirected in the opposite direction.

To analyse this behaviour for fog, we examined the scattering phase function for both ice crystals and water droplets at a wavelength of 3.1 μ, which lies in the near-infrared region. This wavelength is particularly relevant for radiative heating. Our results showed that for both ice and water particles, scattering is strongly forward-directed at this wavelength.

This forward scattering characteristic is beneficial in the context of beam penetration and fog clearance, as it allows the radiation to propagate deeper into the medium without being significantly redirected or reflected. Consequently, more energy can be deposited within the fog layer, leading to enhanced radiative heating and potentially faster fog dissipation.



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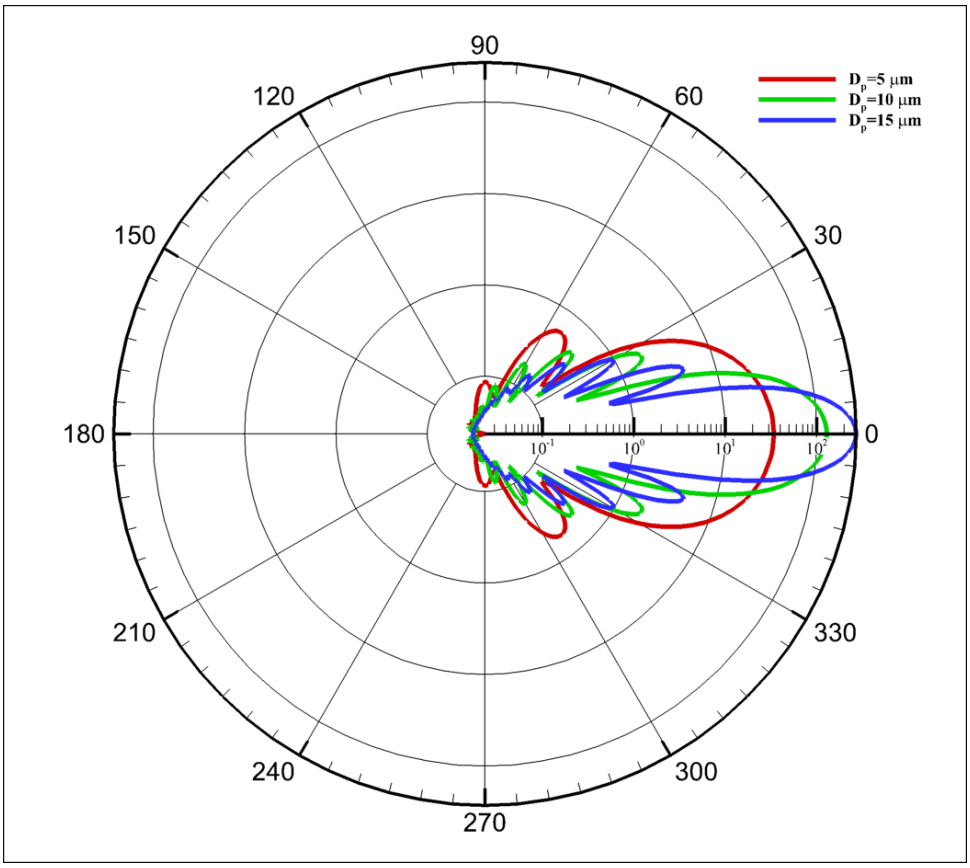

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Figure 4: scattering phase function for various diameters for wavelength 3.1μ: (a) Ice crystals (b) Water droplets

4 Luminous Contours:

In dense fog, the incident radiation encounters a high concentration of suspended water droplets and ice crystals, leading to significant radiation beam obstruction. As a result, the radiation must pass through many droplets and crystals, leading to increased absorption and scattering. In contrast, interactions between the radiation and the particles are reduced under less dense fog conditions, resulting in lower absorption and scattering.

As the radiation transfers its energy to the fog particles, the absorbed energy increases the kinetic energy of the water molecules within the droplets and ice crystals. This kinetic energy increase can raise the particles' local temperature, eventually leading to the water droplets' evaporation and the ice crystals' sublimation or melting. This process effectively reduces the local fog density, creating a tunnel of improved visibility through the fog.

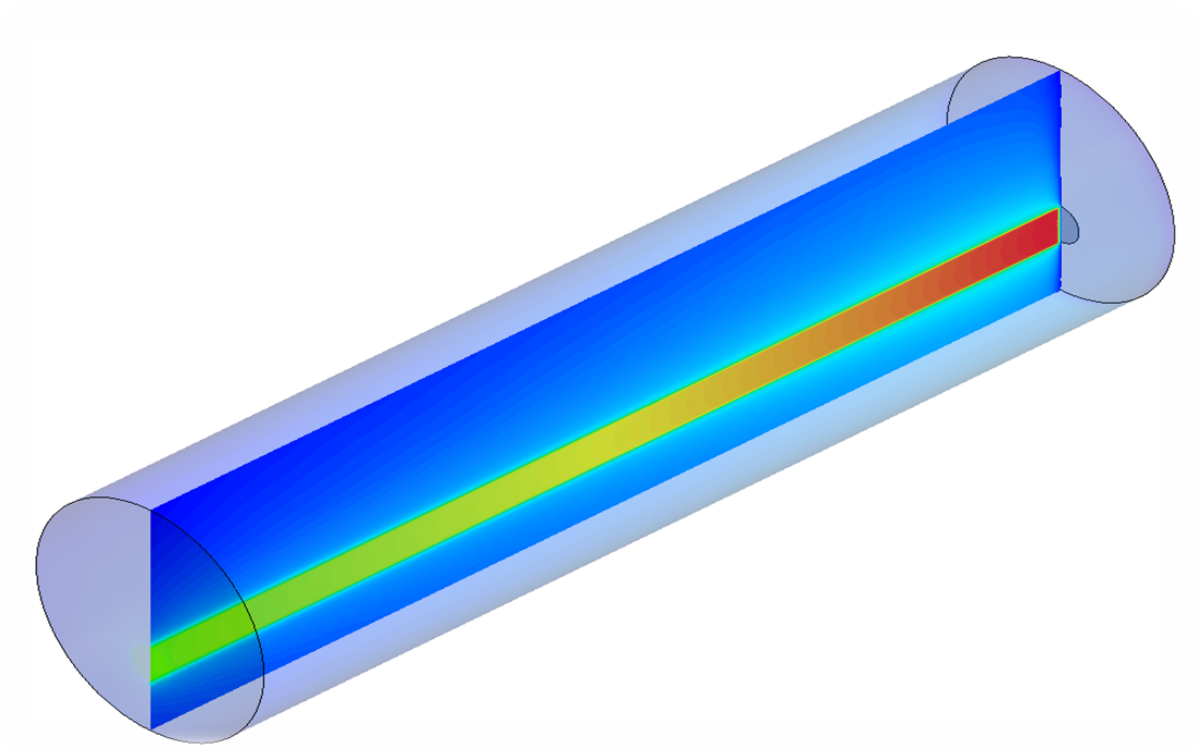
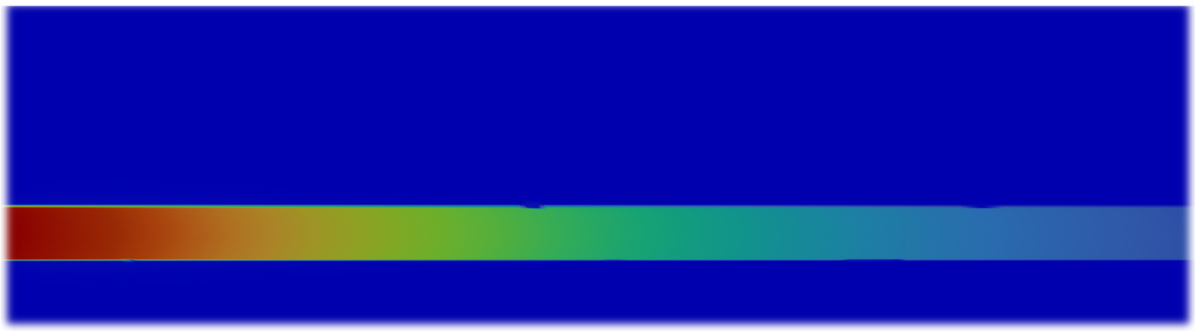


Figure 5: Domain is filled with fog (humid air, water droplets & ice crystals)



(a)

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Figure 6: Lumination Contour: (a) Dense Fog (b) Light Fog

5 Conclusions

1. A collimated radiation beam can clear fog by heating and evaporating water droplets and melting ice crystals.
2. Water droplets in fog typically range from 1–20 μm, while ice crystals range from 20–50 μm in diameter.
3. The infrared range (1–10 μm) shows high absorption for both water and ice, especially ice, making it effective for radiative heating and melting.
4. In this infrared range, the size parameter $x = \frac{2\pi r}{\lambda} > 1$, so Mie theory is valid.
5. The microwave range (1000–10,000 μm) exhibits strong absorption in water, especially at the higher end, due to dielectric heating.
6. For both water and ice in the microwave range, the size parameter $x \ll 1$, indicating the Rayleigh regime, where scattering is weak but absorption remains significant.
7. Analysis of the scattering phase function at 3.1 μm (near-infrared) reveals that both water droplets and ice crystals exhibit strongly forward-directed scattering.
8. Forward scattering allows the radiation beam to penetrate deeper into the fog, improving energy delivery and enhancing the fog-clearing effect.
9. Simulations using the Volume of Fluid (VOF) method and Monte Carlo radiation modelling effectively capture the fog-clearing process.
10. In dense fog, higher absorption and scattering occur, but radiation still deposits energy, leading to evaporation/sublimation and creating a transparent tunnel.
11. Targeting the near-infrared range (1–10 μm) is most effective for fog clearance, mainly due to high absorption by ice.

12. The microwave range (1000–10,000 μm) remains a relevant area for further study.

References

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